SoundSight

Sound Sensing and Pan–Tilt Tracking for Meetings

Team members: to confirm

Affiliation: to confirm

Supervisor: to confirm

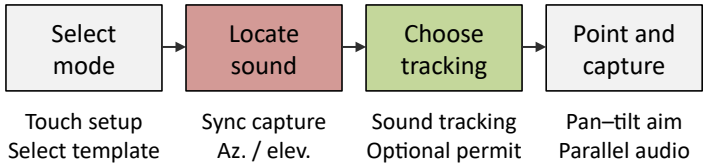
Logo pending

Device photograph
Original pending

AMD Zynq-7020 | PL + PS
Linux / PYNQ

INTRODUCTION

As people move and take turns speaking in meetings, fixed-direction devices need re-aiming. SoundSight targets meeting applications, with future use in robotics and embodied AI for sound direction and speaker identity.

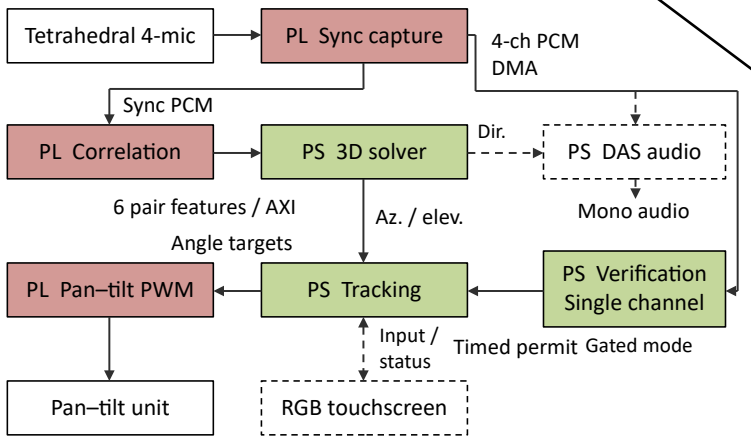


Typical workflow (final design)

Zynq-7020 combines sound sensing and pointing. PL handles synchronous capture, frequency-domain correlation and pan–tilt PWM; PS handles 3D estimation, speaker verification and tracking. The final design adds local RGB touch control and direction-guided audio.

Key technical challenges and design choices

3D direction: Non-coplanar mics constrain azimuth and elevation; no distance estimate.
Scheduling: Direction updates continue during speaker scoring. Passing results grant a timed permit.
Motion: Direction quality, limits and mode gate motion. Axes default off, with stop and timeout guards.
Integration: Directional audio uses PCM and direction; touch links settings and status.



Dashed paths await system integration; DAS has a PC reference.
Speaker verification uses one channel, not DAS output.

System architecture and innovations

CREATIVE DESIGN

Tetrahedral 3D sound localization

Four non-coplanar, tetrahedral mics provide six time-delay constraints. PL extracts correlation features; PS uses subsample interpolation, least squares and filtering to estimate azimuth and elevation, guiding pan–tilt aiming and directional audio.

Asynchronous speaker-gated tracking

A passing score grants a timed permit, so pan–tilt updates need not await each inference. This uses the current direction without guaranteed multi-speaker identity binding.

RESULT

Joint localization, verification and pan–tilt operation

Metric	Run 1	Run 2
Logged duration / s	32.159	27.254
Localization frames read	3012	2552
Speaker evaluations before STOP	6	6
Pan–tilt target commands	117	94
Runtime fault value	0	0
Old localization results dropped	0	1

48 kHz; direction window: 3 frames, at least 1 valid.
Level threshold: 12000 (PCM statistic).

Both board trials combined localization reads, speaker scoring and pan–tilt commands with runtime fault values of 0. The user observed turning toward sound. This supports joint operation on one platform; commands do not confirm that the mechanism reached each target.

Both axes were enabled. The trials used one target template, threshold 0.45 and a 10 s permit. Each run completed 6 speaker evaluations before STOP as direction updates continued. Run 2 dropped 1 old localization result. Angular error, verification accuracy and long-term stability remain unverified.

Next: quantify direction/pointing error; test other speakers, real-mic beamforming and touch; pursue robotic hearing.